



2nd Workshop on Inter-networking challenges for AI, collocated with CoNEXT'26

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Utrecht,
The Netherlands



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Call for papers

AI systems are increasingly distributed across datacenters, wide-area networks, edge platforms, and independently operated domains. Large-scale training pushes beyond the limits of single-site infrastructure; inference and fine-tuning require low-latency, resource-aware placement across heterogeneous compute environments; and agentic AI introduces new patterns of dynamic, multi-party, and often uncoordinated communication.

These developments raise fundamental networking questions. How should networks support AI workloads whose communication patterns differ from traditional cloud traffic? What abstractions, protocols, and measurement methods are needed for wide-area collective communication, distributed inference, edge-core execution, and decentralized AI services? How can such systems provide security, privacy, accountability, and operational trust across administrative boundaries?

The 2nd INet4AI Workshop at ACM CoNEXT'26 invites researchers and practitioners from networking, distributed systems, machine learning systems, and AI infrastructure to explore how AI changes the Internet — and how Internet architecture, protocols, and measurement can enable scalable, trustworthy, and efficient distributed AI.

Topics of interest include, but are not limited to:

- Traffic characterization and measurement of distributed AI training, inference, and agentic workloads;
- Wide-area and inter-domain collective communication for AI workloads;
- Transport protocols, congestion control, routing, and load balancing for distributed AI systems;
- In-network processing, aggregation, caching, and computation for large-scale or federated AI;
- Disaggregated and heterogeneous AI training and inference architectures;
- Edge-core distribution, placement, migration, and orchestration of AI workloads;
- Communication-efficient federated, collaborative, and decentralized learning systems;
- Distributed inference systems, including KV-cache distribution, model serving, and multi-node reasoning;
- Security, privacy, authentication, authorization, and accountability in distributed AI communication;
- Decentralized architectures for AI services across administrative domains;
- Agentic networking, including naming, identity, discovery, capability routing, and agent-to-agent communication;
- Measurement, benchmarking, reproducibility, and performance evaluation of distributed AI and agent systems;
- Real-world deployments, operational experience, case studies, governance, and regulation.

Submitted papers to the workshop must be at most six (6) pages long, excluding references and appendices, in two-column 10pt ACM format. Authors of accepted submissions are expected to present and discuss their work at the workshop. All submissions will be peer-reviewed, and the review process will be double-blind. Please visit the workshop's website to get more information about the submission procedure:

<https://irtf-net4ai.github.io/inet4ai/cfp.html>